

WOOJOO NA

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EDUCATION

Northeastern University

Sep. 2024 – May 2028 (expected)

Ph.D. in Electrical and Computer Engineering | Advisor: Jennifer Dy

· Research Topic: Mathematical foundations of machine learning, optimal transport, and geometrically structured learning algorithms

Tufts University

Sep. 2021 – Dec. 2023

· M.S. in Computer Science, awarded Dec. 2023 (entered in the Ph.D. program)

· Master's Thesis Advisor: Abiy Tasissa

· Master's Thesis: Novel Ensemble Learning Method with Applications in Weakly Supervised Learning

University of Oxford

Oct. 2016 – July 2019

B.A. (Hons) in Mathematics

RESEARCH INTERESTS

Mathematical foundations of machine learning; optimal transport on covariance and other structured spaces; monotone (descent-guaranteed) learning updates and their optimization; learning from incomplete or noisy supervision; learning theory.

PUBLICATIONS & PREPRINTS

Woojoo Na, Jennifer Dy. "ITSPACE: Monotone Gaussian Optimal Transport Updates."

International Conference on Machine Learning (ICML), 2026. Accepted as a regular paper. OpenReview · arXiv

- A proximal majorization–minimization method for the exact Bures–Wasserstein (Gaussian W_2) objective on covariances, with closed-form updates in a square-root factorization; each iteration satisfies a sufficient-decrease inequality in exact arithmetic, with a certificate-gap bound under inexact polar computations.

- First author and sole student author; led problem formulation, theoretical development, algorithm design, experiments, and writing.

Woojoo Na, Abiy Tasissa. "RACH-Space: Reconstructing Adaptive Convex Hull Space with Applications in Weak Supervision."

arXiv preprint, 2023. arXiv

- First author and sole student author; introduced an adaptive convex-hull reconstruction method for weak supervision.

- Reported state-of-the-art performance on the weak-supervision benchmark tasks considered at the time.

RESEARCH EXPERIENCE

Northeastern University

Sep. 2024 – Present

Graduate Researcher, Advisor: Professor Jennifer Dy

- Developed *ITSPACE*, a proximal majorization–minimization method for Gaussian (Bures–Wasserstein) optimal transport between covariances: closed-form updates in a square-root factorization that preserve positive semidefiniteness and support rank-restricted factors (ICML 2026).

- Proved a per-iteration sufficient-decrease guarantee for the BW objective in exact arithmetic and a certificate-gap bound under inexact polar computations; on real-world covariance-alignment benchmarks, *ITSPACE* reaches low-BW-gap solutions substantially faster than BW gradient descent and entropically regularized sample-OT baselines.

- Led the project end-to-end (problem formulation, theory, algorithm design, experiments, manuscript). Current work: monotone transport-based learning updates and computationally light, mathematically principled adaptation algorithms.

Data Intensive Studies Center, Tufts University

Sep. 2021 – Dec. 2023

Graduate Researcher, Advisor: Professor Abiy Tasissa

- Developed *RACH-Space*, an adaptive convex-hull reconstruction algorithm for weak supervision under partial and noisy label information.
- Built the theoretical motivation, implemented the method, and evaluated performance against weak-supervision baselines; reported state-of-the-art results on the benchmark tasks considered at the time.
- Produced a master's thesis and arXiv preprint on weakly supervised learning and ensemble methods.

University of Oxford, Mathematical Institute

July 2019 – Dec. 2019

Undergraduate Researcher, Advisor: Professor Christophe Petit

- Developed a group-theoretic trapdoor attack against Cayley hash function parameters proposed at the NutMiC 2019 conference.
- Presented the work at the 17th IMA International Conference on Cryptography and Coding. [Conference link](#)

University of Oxford, Department of Computer Science

July 2019 – Dec. 2019

Undergraduate Researcher, Advisor: Professor Thomas Lukasiewicz

- Studied similarities between transformer encoder word embeddings and conceptual-space geometry.
- Visualized embedding structure to investigate low-dimensional features shared by words in the same semantic category.

University of Oxford, Mathematical Institute

June 2018 – Oct. 2018

Undergraduate Researcher, Advisor: Dr. Andrey Kormilitzin

- Competed in the n2c2 challenge on adverse drug events and medication extraction from electronic health records.
- Implemented CNN and character-level word-embedding models for concept extraction and relation extraction from clinical text.

University of Oxford, Department of Computer Science

June 2018 – Oct. 2018

Undergraduate Researcher, Advisor: Professor Thomas Lukasiewicz

- Worked on knowledge graph completion for ontology reasoning.

SELECTED TALKS & PRESENTATIONS

“ITSPACE: Monotone Gaussian Optimal Transport Updates”	July 2026
<i>Poster presentation, International Conference on Machine Learning (ICML 2026), Seoul, Korea</i>	
“Novel Ensemble Learning Method with Applications in Weakly Supervised Learning”	Dec. 2023
<i>Master's Thesis Defense, Tufts University</i>	
“Bounding Worst-Case Errors of Weak Supervision Algorithms”	April 2022
<i>Tufts University Mathematics Graduate Seminar</i>	
“Security of Cayley Hash Functions”	Jan. 2020
<i>Invited Speaker, Ewha Womans University Cryptography and Mathematics Seminar</i>	
“Trapdoor Attack on Cayley Hash Function Parameters Proposed at NutMiC 2019 Conference”	Dec. 2019
<i>17th IMA International Conference on Cryptography and Coding Conference link</i>	
“A Deep Learning Approach to Concept Extraction from Medical Texts”	Oct. 2018
<i>Poster Session, University of Oxford Mathematical Institute</i>	

HONORS & AWARDS

- Tufts University Linda M. Abriola Graduate Fellowship (\$5,000; awarded to 2 graduate students)	2022–2023
- Tufts University Linda M. Abriola Graduate Fellowship (\$5,000; awarded to 2 graduate students)	2021–2022
- Brigade Commander Award, Republic of Korea Army Start-up Competition	2021
- Lady Margaret Hall Academic Development Fund for Undergraduate Research (£500)	2018

TEACHING EXPERIENCE

Tufts University, Department of Computer Science

Jan. 2023 – May 2023

Teaching Assistant, CS170 Computation Theory

ADDITIONAL EXPERIENCE

Republic of Korea Army, 2nd Artillery Brigade

Feb. 2020 – Aug. 2021

Sergeant, Fire Direction Operator

- Completed mandatory military service; received three brigade-level awards for excellence and was selected for combined command-post training with the U.S. Army and Republic of Korea Army.

Utpal Shanghvi Global School, Mumbai, India

July 2017 – Aug. 2017

Project Intern

- Worked on economic sustainability and public-health projects for villagers in Wada, Maharashtra, India.

LEADERSHIP & SERVICE

Oxford Invariants (Oxford Mathematics Society)

April 2018 – April 2019

Treasurer

Oxford Mathematical Institute, Problem Solving Matters Program

June 2018 – Oct. 2018

Mentor for Underprivileged High School Students

St. Nicholas Primary School, Oxford, UK

Oct. 2016 – April 2018

Volunteer Mathematics Teacher for Immigrant Primary Students

TECHNICAL SKILLS & MATHEMATICAL BACKGROUND

Programming

Python, C++, Lean

Mathematics

Optimal transport, optimization, probability, linear algebra, topology, group theory, algebraic geometry